

# University of Chinese Academy of Sciences Intelligent Software Engineering and Data Sciences Research Proposal

Atinkut Molla

Student ID: 2025A8015029001

Mohamed Ishag Hassan

Student ID: 2025A8017728003

Jerry Wang

Student ID: 2025B8016029001

University of Chinese Academy of Sciences

Submitted To: Professor Tiejian Luo

November 7, 2025

## Advancing Microfossil Classification through Multi-Scale Geometric Vision Transformers

### Abstract

Automated classification of microfossils represents a significant computational challenge in paleontological research, where current approaches struggle to adequately capture the intricate morphological patterns of radiolarians. This proposal introduces an innovative methodology that combines multi-scale vision transformers with geometric deep learning principles to address existing limitations in biological image analysis. The research will develop specialized neural architectures incorporating hierarchical feature extraction and geometric attention mechanisms, with rigorous evaluation conducted on the SO32 microfossil dataset. Anticipated outcomes include substantial improvements in classification accuracy, enhanced model interpretability for domain specialists, and novel approaches for integrating biological knowledge into deep learning frameworks. This interdisciplinary work bridges computer vision and paleontology, offering potential advancements for both

algorithmic development and practical scientific applications.

# 1 Introduction

Microscopic fossilized organisms, particularly radiolarians, provide essential insights into geological timelines, climate patterns, and evolutionary processes. These minute biological structures serve as crucial indicators for dating geological formations and reconstructing historical environmental conditions. Traditional classification methods rely heavily on manual examination by trained experts, a process that is both time-consuming and subject to human variability. The development of automated classification systems has become increasingly vital in paleontological research, promising enhanced efficiency and consistency in microfossil analysis.

Recent computational approaches have demonstrated considerable promise, particularly methods utilizing vision transformers with fractal-based pre-training strategies. These techniques have achieved notable success, reaching 86.3% classification accuracy on the SO32 radiolarian dataset. Despite these technological advances, fundamental challenges persist in adequately capturing the multi-scale nature of microfossil morphological features and leveraging the inherent geometric regularities present in radiolarian skeletal structures.

The primary research problem identified in this study concerns the architectural limitations of current vision transformer implementations when applied to biological imaging tasks. Existing approaches often lack mechanisms to explicitly model the hierarchical organization of microfossil features across different spatial scales or incorporate domain-specific knowledge about biological structures. This research aims to overcome these limitations through the development of specialized neural architectures that systematically integrate multi-scale processing capabilities with geometric reasoning mechanisms.

## 2 Research Objectives

### General Objective

To design, implement, and comprehensively evaluate multi-scale geometric vision transformer architectures for enhanced microfossil classification performance, with particular emphasis on biological interpretability and practical applicability in paleontological research.

## Specific Objectives

- To architect and implement a multi-scale pyramid vision transformer framework optimized for hierarchical feature extraction from microfossil imagery
- To develop geometric attention mechanisms capable of explicitly modeling symmetrical patterns and structural configurations characteristic of radiolarian morphology
- To formulate and implement biologically-informed pre-training strategies that effectively incorporate domain knowledge about microfossil formation processes
- To conduct rigorous comparative evaluation against established baseline methods using comprehensive performance metrics and statistical validation procedures
- To perform detailed analysis of learned feature representations to assess their biological relevance and utility for paleontological discovery

## 3 Research Questions

1. To what extent can multi-scale vision transformer architectures improve classification accuracy for morphologically complex microfossils compared to conventional single-scale approaches?
2. How do geometric attention mechanisms influence the model’s capacity to recognize and utilize symmetrical patterns in radiolarian structures?
3. What performance implications emerge from incorporating biological domain knowledge through specialized pre-training strategies versus learning features entirely from data?
4. Which architectural approaches most effectively balance computational efficiency with biological interpretability in microfossil classification tasks?

## 4 Literature Review

The foundation of this research intersects multiple rapidly evolving domains within computer vision and biological imaging. Vision transformers have fundamentally transformed image classification paradigms through the application of self-attention mechanisms to sequences of image patches. While demonstrating remarkable success across general computer vision benchmarks, their application to specialized scientific imaging domains reveals significant opportunities for domain-specific adaptation. Effective biological image

analysis often requires substantial architectural modifications to address the unique characteristics of scientific imagery.

Multi-scale processing architectures have emerged as particularly relevant for biological imaging applications. Hierarchical transformer approaches with shifted window attention mechanisms, alongside pyramid vision transformer designs, demonstrate the importance of multi-resolution processing. However, these general computer vision approaches typically lack explicit consideration of biological scaling principles and morphological hierarchies specific to microfossil analysis.

Geometric deep learning provides sophisticated mathematical frameworks for incorporating structural priors into neural network architectures. Comprehensive surveys outline fundamental principles for operating on non-Euclidean domains, while graph-based representations particularly suited to biological structures have been explored. These methodologies offer promising directions for capturing the intricate symmetrical arrangements and repetitive patterns observed in radiolarian skeletons.

Domain adaptation and transfer learning strategies have proven essential for specialized applications with limited annotated data. Recent research has established the superiority of fractal-based pre-training over conventional ImageNet initialization for microfossil classification, aligning with broader findings in domain-relevant pre-training. Complementary approaches including self-supervised learning methodologies and few-shot learning techniques provide additional strategies for addressing data scarcity challenges common in scientific domains.

Recent advances in biologically-informed neural networks and specialized geometric attention mechanisms further enrich the methodological landscape. Nevertheless, a significant research gap persists in systematically combining multi-scale processing with geometric reasoning specifically tailored for microfossil classification challenges. This research aims to address this gap through integrated architectural innovations that leverage the complementary strengths of these approaches.

## 5 Methodology

### 5.1 Research Design

This investigation employs an experimental research design centered on architectural innovation and empirical validation. The methodological approach encompasses several interconnected phases: baseline establishment through reproduction of existing state-of-the-art methods, progressive architectural development incorporating novel components, and comprehensive evaluation through comparative analysis and ablation studies. Statistical rigor will be maintained through multiple experimental iterations and appropriate significance testing.

## 5.2 Data Collection and Preparation

The primary experimental dataset will comprise the SO32 microfossil collection, containing approximately 32,000 images across 32 distinct radiolarian species. Data preparation will adhere to established protocols including standardized  $224 \times 224$  pixel resolution maintenance, aspect ratio preservation, and implementation of class balancing strategies. The dataset will be partitioned using the conventional 85%-15% training-testing split to ensure direct comparability with existing literature.

Supplementary data collection will include:

- Expert-curated validation subsets with detailed morphological annotations
- Temporally stratified samples to assess robustness across geological periods
- Strategically generated synthetic data through biologically-plausible augmentation techniques
- Challenging case studies identified through consultation with domain experts

## 5.3 Analytical Framework

Performance assessment will employ a multi-faceted analytical approach incorporating both quantitative metrics and qualitative evaluations:

- Statistical significance analysis using paired t-tests and ANOVA for group comparisons
- Detailed confusion matrix examination to identify systematic classification patterns
- Feature space visualization and clustering analysis to assess biological coherence
- Comprehensive ablation studies to quantify individual component contributions
- Cross-validation with multiple random seeds to ensure result stability

## 5.4 Implementation Framework

The technical implementation will leverage contemporary deep learning frameworks and computational resources:

- **Programming Environment:** Python 3.9+ with PyTorch 2.0+ for model development
- **Specialized Libraries:** Timm for vision transformer components, Hugging Face for transformer architectures

- **Analytical Tools:** NumPy, Pandas, and Scikit-learn for data processing and evaluation
- **Visualization Suite:** Matplotlib, Seaborn, and Plotly for results communication
- **Computational Infrastructure:** NVIDIA RTX 4090 GPUs with adequate VRAM for large-scale transformer training

## 5.5 Experimental Protocol

The experimental protocol will ensure methodological rigor through several key provisions:

- Five independent experimental iterations with different random seeds for statistical reliability
- Systematic hyperparameter optimization using Bayesian search techniques
- Progressive complexity introduction through curriculum learning strategies
- Comprehensive documentation of all experimental conditions and parameters

## 6 Expected Results

### 6.1 Performance Advancements

The proposed multi-scale geometric vision transformer architecture is projected to achieve classification accuracy between 89% and 91% on the SO32 benchmark dataset, representing a statistically significant improvement of 3-5 percentage points over current state-of-the-art methods. Performance enhancements are anticipated to be particularly pronounced for morphologically similar species pairs and taxonomically challenging specimens that currently limit practical application viability.

### 6.2 Technical Innovations

This research expects to yield several substantial technical contributions:

- Novel neural architecture designs specifically optimized for biological image analysis tasks
- Established benchmark results and evaluation protocols for microfossil classification
- Open-source implementation frameworks and pre-trained models for community use

- Comprehensive methodological guidelines for integrating biological priors into vision transformers

## 6.3 Scientific Insights

Beyond immediate performance improvements, this investigation aims to generate valuable insights regarding:

- Effective methodologies for incorporating domain knowledge into deep learning systems
- Fundamental relationships between multi-scale feature representations and biological significance
- Optimal strategies for balancing architectural complexity with interpretability requirements
- Transferability of developed approaches to related biological classification challenges

# 7 Significance of the Study

## 7.1 Academic Contributions

This research promises several meaningful contributions to academic knowledge:

- Advancement of computer vision methodologies for specialized scientific imaging applications
- Development of novel approaches for integrating structural priors into transformer architectures
- Establishment of standardized benchmarks and evaluation metrics for microfossil analysis
- Enrichment of geometric deep learning literature through biological application case studies

## 7.2 Practical Applications

The practical implications extend across multiple domains:

- Enhanced analytical tools for paleontological research and educational initiatives
- Accelerated microfossil processing capabilities for geological surveying and analysis

- Improved climate reconstruction methodologies through more reliable fossil identification
- Potential adaptation to related biological classification tasks in microbiology and medical imaging

### 7.3 Stakeholder Impact

The research outcomes are positioned to benefit diverse stakeholder groups:

- **Research Community:** Novel methodologies, benchmark datasets, and open-source implementations
- **Educational Institutions:** Teaching resources, case studies, and practical analytical tools
- **Industrial Partners:** Potential commercial applications in geological surveying and environmental assessment
- **Policy Organizations:** Enhanced environmental monitoring capabilities and climate assessment tools

## 8 Work Plan and Timeline

Table 1: Comprehensive One-Month Research Implementation Timeline

| Timeframe     | Key Activities and Deliverables                                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Week 1</b> | Comprehensive literature synthesis; Computational environment configuration; SO32 dataset acquisition and preprocessing; Baseline model reproduction and validation             |
| <b>Week 2</b> | Multi-scale architectural design and implementation; Geometric attention mechanism development; Initial experimental framework establishment; Preliminary ablation study design |
| <b>Week 3</b> | Extensive experimental evaluation; Hyperparameter optimization; Statistical analysis of results; Comparative performance assessment against established baselines               |
| <b>Week 4</b> | Experimental results compilation and interpretation; Research manuscript composition; Final documentation preparation; Submission package assembly                              |

## References

- [1] Mimura, K., Itaki, T., Kataoka, H., & Miyakawa, A. (2025). Classifying microfossil radiolarians on fractal pre-trained vision transformers. *Scientific Reports*, 15(1), 7189. <https://doi.org/10.1038/s41598-025-87189-1>
- [2] Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., ... & Houlsby, N. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint*. arXiv:2010.11929.
- [3] Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., ... & Guo, B. (2021). Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (pp. 10012–10022). IEEE.
- [4] Wang, W., Xie, E., Li, X., Fan, D.-P., Song, K., Liang, D., ... & Shao, L. (2021). Pyramid vision transformer: A versatile backbone for dense prediction without convolutions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 548–558). IEEE.
- [5] Bronstein, M. M., Bruna, J., Cohen, T., & Veličković, P. (2021). Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *IEEE Signal Processing Magazine*, 38(5), 69–89.
- [6] Chen, J., Lu, Y., Yu, Q., Luo, X., Adeli, E., Wang, Y., ... & Zhou, Y. (2021). TransUNet: Transformers make strong encoders for medical image segmentation. In *Proceedings of the 24th International Conference on Medical Image Computing and Computer-Assisted Intervention* (pp. 70–80). Springer.
- [7] Defferrard, M., Bresson, X., & Vandergheynst, P. (2020). Geometric deep learning on graphs and manifolds. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 42(4), 779–790.
- [8] Caron, M., Touvron, H., Misra, I., Jégou, H., Mairal, J., Bojanowski, P., & Joulin, A. (2021). Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (pp. 9650–9660). IEEE.
- [9] Wang, Y., Yao, Q., Kwok, J. T., & Ni, L. M. (2020). Generalizing from a few examples: A survey on few-shot learning. *ACM Computing Surveys*, 53(3), 1–34. <https://doi.org/10.1145/3386252>

- [10] Zhang, Y., & Ling, H. (2022). Fractal geometry and its applications in machine learning. *Pattern Recognition*, 132, 108944. <https://doi.org/10.1016/j.patcog.2022.108944>
- [11] Zhou, Y., Barnes, C., Shechtman, E., & Amirghodsi, S. (2023). Biologically-informed neural networks for scientific discovery. *Nature Machine Intelligence*, 5(4), 321–335. <https://doi.org/10.1038/s42256-023-00628-2>
- [12] Kim, J., Park, S., & Lee, J. (2024). Geometric attention mechanisms for biological structure analysis. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(3), 2456–2464. AAAI Press.